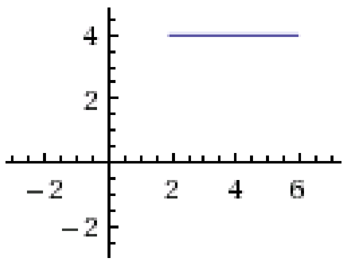
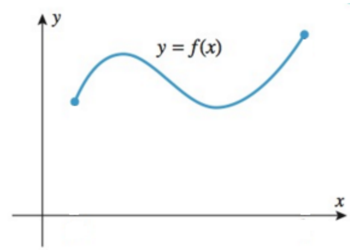
MTH 201 Chapter 1: Application of Integration

1.4 Length of a Plane Curve

General Idea:
Determine the length of the following curves:



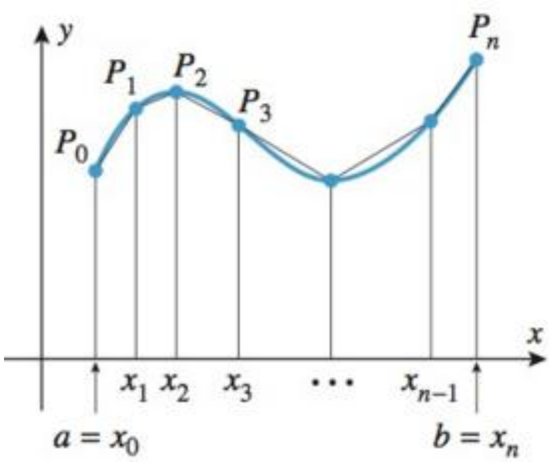


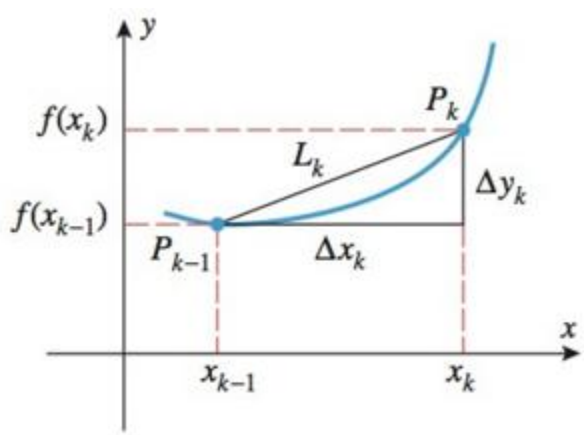
Here the length of the curve is easy to calculate.
$Length = 6 - 2 = 4$

Here the length of curve seems almost impossible to calculate!

The Specifics:
Suppose $f(x)$ is a function with a continuous first derivative on the interval $[a,b]$. Also suppose that a curve C is defined by the equation $y = f(x)$ where $a \le x \le b$.

Now, we wish to find the length of the curve C. To do this we first get an approximation by dividing $[a,b]$ into n equally spaced subintervals (with endpoints $x_0, x_1, ..., \& x_n$) and approximating the curve with line segments. A picture is shown below.





From the above picture we see that we let P_k have coordinates $(x_k, f(x_k))$. Note that the length of the kth line segment is:

$$L_k = \sqrt{(\Delta x)^2 + (\Delta y_k)^2} = \sqrt{(\Delta x)^2 + (f(x_k) - f(x_{k-1}))^2}$$

We can see that the approximation gets better as n increases. This leads to the following definition:
The length of the curve C is:

$$L = \lim_{n\to\infty} \sum_{k=1}^{n} L_k = \lim_{n\to\infty} \sum_{k=1}^{n} \sqrt{(\Delta x)^2 + (f(x_k) - f(x_{k-1}))^2}$$

Now, by the mean value theorem we know that there is a number x_k^* in (x_{k-1}, x_k) such that

$$\frac{f(x_k)-f(x_{k-1})}{x_k - x_{k-1}} = f'(x_k^*) \quad \Rightarrow \quad f(x_k)-f(x_{k-1}) = f'(x_k^*)(x_k - x_{k-1}) \quad \Rightarrow \quad f(x_k)-f(x_{k-1}) = f'(x_k^*)\Delta x$$

Using this and the fact that $\Delta x > 0$ we see that:

$$\sqrt{(\Delta x)^2 + \left(f(x_k)-f(x_{k-1})\right)^2} = \sqrt{(\Delta x)^2 + \left(f'(x_k^*)\Delta x\right)^2} = \left(\sqrt{1+\left(f'(x_k^*)\right)}\right)\Delta x$$

Now, by definition we know that the length of the curve C is given by:

$$L = \lim_{n\to\infty}\sum_{k=1}^{n} L_k = \lim_{n\to\infty}\sum_{k=1}^{n}\left(\sqrt{1+\left(f'(x_k^*)\right)^2}\right)\Delta x$$

Now, we notice that the limit on the right side is a definite integral! In particular,

$$L = \lim_{n\to\infty}\sum_{k=1}^{n}\left(\sqrt{1+\left(f'(x_k^*)\right)^2}\right)\Delta x = \int_a^b \sqrt{1+\left(f'(x)\right)^2}\,dx$$

Now we have derived the following formula.

Theorem (The Arc Length Formula): If $f'(x)$ is continuous on $[a,b]$, then the length of the curve $y = f(x)$ where $a \le x \le b$ is

$$L = \int_a^b \sqrt{1+\left[f'(x)\right]^2}\,dx = \int_a^b \sqrt{1+\left[\frac{dy}{dx}\right]^2}\,dx$$

Note: If a curve has the equation $x = g(y)$, where $c \le y \le d$ and $g'(y)$ is continuous on $[c,d]$ then

$$L = \int_c^d \sqrt{1+\left[g'(y)\right]^2}\,dy = \int_c^d \sqrt{1+\left[\frac{dx}{dy}\right]^2}\,dy$$

Example 1: Find the exact length of the curve determined by $y = x^{2/3}$ between the points $(1,1)$ and $(8,4)$.

a) Determine this by integrating with respect to *x*.

We see that $\frac{dy}{dx} = \frac{2}{3}x^{-1/3}$ is continuous on the interval $[1,8]$.

Setup:

In this case we see that $1 \le x \le 8$ and $y = x^{2/3}$, so $\frac{dy}{dx} = \frac{2}{3}x^{-1/3}$. So the desired length is given by:

$$L = \int_1^8 \sqrt{1+\left(\frac{2}{3}x^{-1/3}\right)^2}\,dx = \int_1^8 \sqrt{1+\frac{4}{9x^{2/3}}}dx = \int_1^8 \sqrt{\frac{1}{9x^{2/3}}\left(9x^{2/3}+4\right)}dx = \int_1^8 \frac{1}{3x^{1/3}}\sqrt{\left(9x^{2/3}+4\right)}dx$$

Evaluate:

Let $u = 9x^{2/3}+4$ and thus $du = 6x^{-1/3}dx$, so $\frac{1}{6}du = x^{-1/3}dx$.

Our limits of integration become: $u(1) = 9(1)^{2/3}+4 = 13$ & $u(8) = 9(8)^{2/3}+4 = 40$.

$$\int_1^8 \frac{1}{3x^{1/3}}\sqrt{\left(9x^{2/3}+4\right)}dx = \int_{13}^{40} \frac{1}{18}\sqrt{u}\,du = \left(\frac{1}{27}u^{3/2}\right)\Bigg|_{13}^{40} = \frac{1}{27}\left(40^{3/2}-13^{3/2}\right)$$

b) Determine this arc length by integrating with respect to *y*.

We see that $x = y^{3/2}$ and $\frac{dx}{dy} = \frac{3}{2}y^{1/2}$ is continuous on the interval $[1,4]$.

Setup:

In this case we see that $0 \le y \le 4$ and $x = y^{3/2}$, so $\frac{dx}{dy} = \frac{3}{2}y^{1/2}$. So the desired length is given by:

$$L = \int_1^4 \sqrt{1+\left(\frac{3}{2}y^{1/2}\right)^2}\,dy = \int_1^4 \sqrt{1+\frac{9}{4}y}\,dy$$

Evaluate:

Let $u = 1+\frac{9}{4}y$ and thus $du = \frac{9}{4}dy \rightarrow \frac{4}{9}du = dy$.

Our limits of integration become: $u(1) = 1+\frac{9}{4}(1) = \frac{13}{4}$ & $u(4) = 1+9 = 10$

$$L = \int_1^4 \sqrt{1+\frac{9}{4}y}\,dy = \frac{4}{9}\int_{\frac{13}{4}}^{10} \sqrt{u}\,du = \frac{4}{9}\left(\frac{2}{3}u^{3/2}\right)\Bigg|_{\frac{13}{4}}^{10} = \frac{8}{27}\left(10^{3/2}-\left(\frac{13}{4}\right)^{3/2}\right) = \frac{4^{3/2}}{27}\left(10^{3/2}-\left(\frac{13}{4}\right)^{3/2}\right) = \frac{1}{27}\left(40^{3/2}-13^{3/2}\right)$$

Example 2: Set up, and evaluate using your calculator, an integral equal to the arc length of the arc of the parabola $\frac{1}{2}y^2+1=x$ from $(1,0)$ to $(3,2)$.

We see that $\frac{dx}{dy}=y$ is continuous on the interval $[0,2]$.

Setup:

In this case we see that $0\le y\le 2$ and $\frac{dx}{dy}=y$. So the desired length is given by:

$$L=\int_0^2\sqrt{1+y^2}\,dy\approx 2.96$$

Note: we will learn to evaluate integrals like this in Chapter 3.

Definition: Suppose the curve C is given by $y=f(x)$ where $a\le x\le b$ and $f'(x)$ is continuous on $[a,b]$. We define the arc length function for the curve C as the function with domain $[a,b]$ given by the formula:

$$s(x)=\int_a^x\sqrt{1+\left(f'(t)\right)^2}\ dt$$

Note: The point $(a,f(a))$ is often referred to as the starting point, and in some problems we use $[a,\infty)$ as the interval rather than $[a,b]$.

Note: By the fundamental theorem of calculus we see that $\frac{ds}{dx}=\sqrt{1+\left(f'(x)\right)^2}$. So, the differential of arc length is $ds=\sqrt{1+\left(f'(x)\right)^2}\,dx=\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$.

Note: When x is a function of y the arc length differential becomes: $ds=\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$.

Note: When a curve C is given by a function with a continuous first derivative we call C a *smooth curve*. It is important to note that in this section we are working with smooth curves, but the notion of arc length can be generalized to include curves that are given by a continuous function (such a generalization does not even require that the function is differentiable!).

Example 3: Consider the function $f(x) = x^2 - \frac{1}{8}\ln(x)$.

a) Find the arc length function (simplify until no integral remains) for this function $(1, f(1))$ is taken to be the starting point (Hint: the radical should simplify nicely).

By the definition above we know that $s(x) = \int_1^x \sqrt{1+(f'(t))^2}\, dt$. So we compute.

$$s(x) = \int_1^x \sqrt{1+(f'(t))^2}\, dt = \int_1^x \sqrt{1+\left(2t - \frac{1}{8t}\right)^2}\, dt = \int_1^x \sqrt{1+4t^2 - \frac{1}{2} + \frac{1}{64t^2}}\, dt$$

$$= \int_1^x \sqrt{4t^2 + \frac{1}{2} + \frac{1}{64t^2}}\, dt = \int_1^x \sqrt{\left(2t + \frac{1}{8t}\right)^2}\, dt = \int_1^x \left(2t + \frac{1}{8t}\right) dt$$

$$= \left(t^2 + \frac{1}{8}\ln(t)\right)_1^x = x^2 + \frac{1}{8}\ln(x) - 1$$

Thus, the answer is $s(x) = x^2 + \frac{1}{8}\ln(x) - 1$.

b) Use your answer to (a) to find the exact arc length along the curve from $(1, f(1))$ to $(3, f(3))$

We want to find $s(3)$. So using our answer for part (a) we compute: $s(3) = 9 + \frac{\ln(3)}{8} - 1 = 8 + \frac{\ln(3)}{8}$.

Therefore the length of the curve $f(x)$ on $[1,3]$ is exactly $s(3) = 8 + \frac{\ln(3)}{8}$.